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# Data Science Fundamentals

## Dimensionality Reduction

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*Before we get started…*

# Practicalities

# Learning objectives

- You understand and apply different types of regression, know when and how to use them, can execute them and put them in to practice as well as interpret the results.
- You recognize overfitting in regression models and learn techniques to prevent overfitting.
- You learn about different clustering techniques and understand when and how to use them for grouping data.
- You gain an understanding of dimensionality reduction techniques and learn how to apply them to compress and visualize data.

# Course material

- **Canvas**
  - Slides & exercises from classes
- **Online**
  - https://www.w3schools.com/datascience/
- **Software**
  - JupyterLab (or work via Anaconda/VS Code if you prefer)
- **The Internet**



# Course breakdown

- **Course Structure:**
  - Lectures: 3 hours per week (theory and exercises)
  - Attendance: Mostly not compulsory, but strongly recommended.

- **Examination Breakdown:**
  - 30%: Continuous assessment (mandatory exercises)
  - 70%: Final exam (oral, mostly theoretical with some practical elements)

# Is it difficult to pass?

It’s perfectly doable **if you put in the work**

- Come to class
- Participate in the lessons
- Make your exercises



# How to be an ideal student

- **Attend classes** - even when you party hardy
- **Show respect** - towards your instructor and fellow students
- **Be punctual** - or you'll find the door closed
- **Meet deadlines** - late submissions will earn a solid zero
- **Have a question?** - just ask! I don't bite
- **Know it all?** - help your classmates, they might repay you with pizza
- **Do your own work!** - shortcuts will lead to a short academic career ;)

# Syllabus

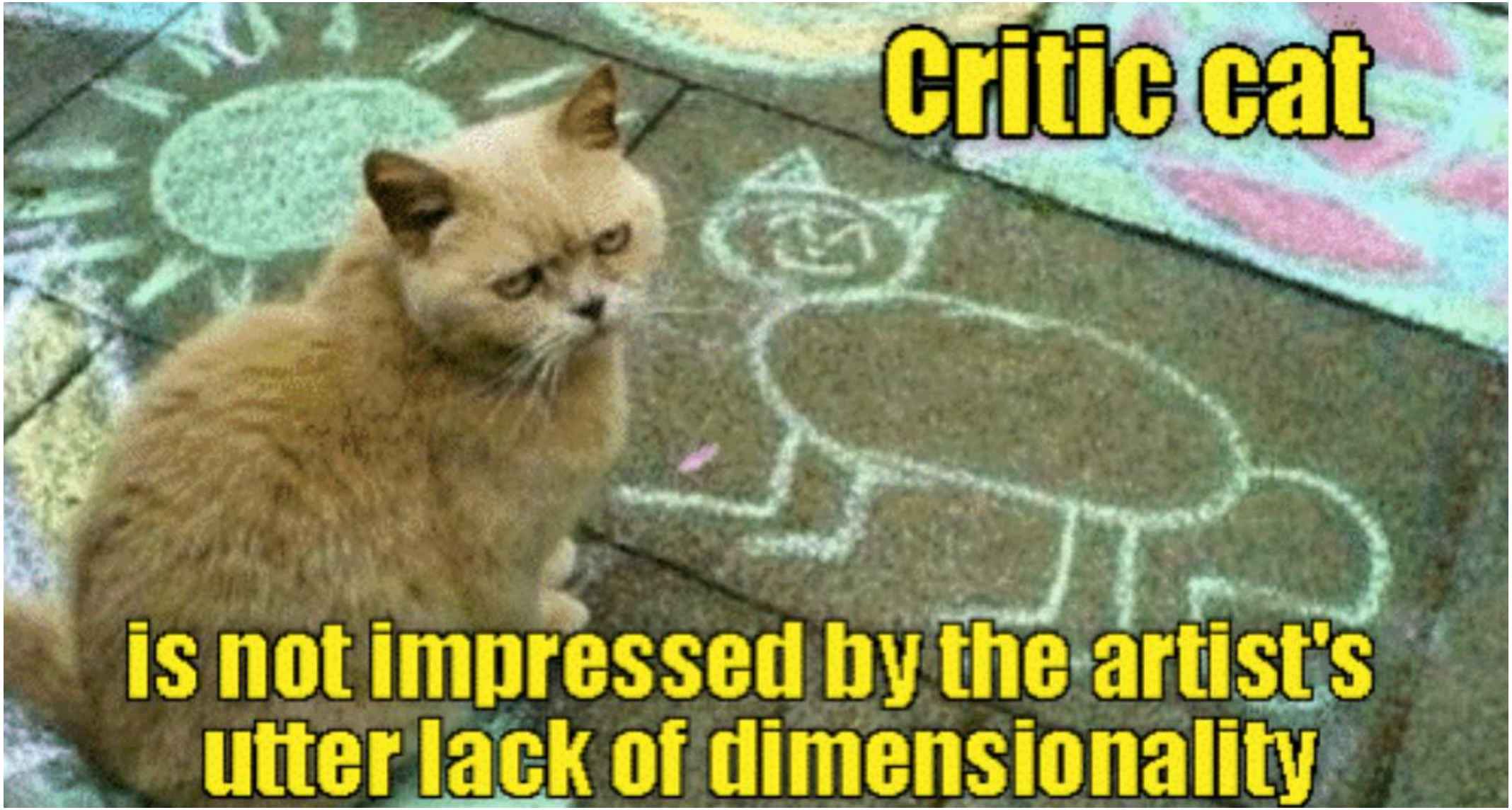
| | | | |
|---|---|---|---|
| Week - 1 | Introduction & Python/JupyterLab Setup | Week - 9 | Easter Holiday |
| Week - 2 | Regression - Linear regression | Week - 10 | Easter Holiday |
| Week - 3 | Regression - Polynomial regression | Week - 11 | Clustering - Hierarchical clustering |
| Week - 4 | Classification - Logistic regression | Week - 12 | Dimensionality reduction: Feature Analysis - PCA |
| Week - 5 | Classification - Decision trees | Week - 13 | Regularization: Lasso regression |
| Week - 6 | Classification - kNN | Week - 14 | Recap: Theory + Excercises |
| Week - 7 | Clustering - kMeans | Week - 15 | Exam Preperation + Q&A |
| Week - 8 | Lifesnaps guidance | | |

(subject to change)

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*Now we get to the fun part*

# Dimensionality Reduction


Critic cat
is not impressed by the artist's
utter lack of dimensionality

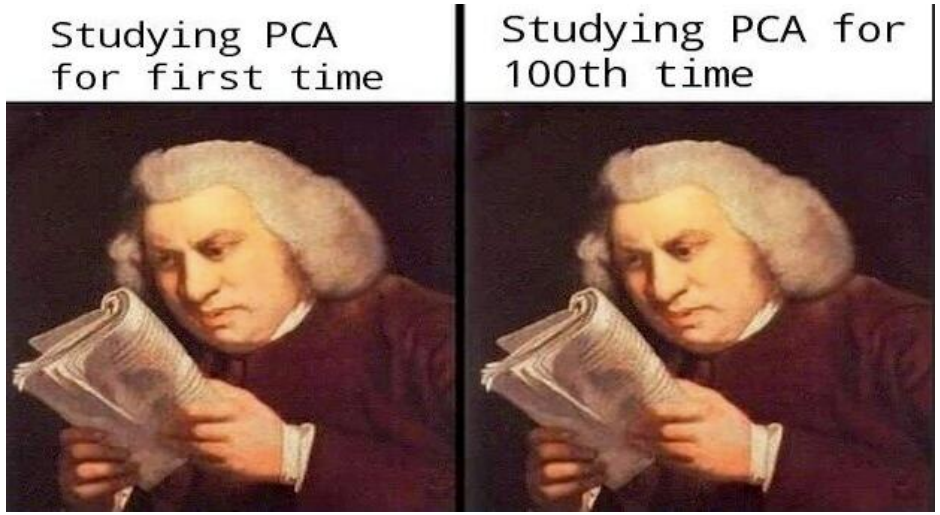
# Dimensionality Reduction

- Sometimes our data is too big to store/to explore manually
- Ideally, we want to keep all data points
- Solution: **reduce the amount of variables**
  - = dimensionality reduction
- **Unsupervised** learning process
- Reduce the amount of variables in a dataset
- Ideal scenario: **no** information loss
- Realistic scenario: **minimal** information loss



# PCA

- ‘Principal Component Analysis’: reduce the variables to a few ‘principal components’
- Only one of the many dimensionality reduction techniques available
- Bundle variables with **high correlations** together
- Try to keep as much explained variance in the dataset as possible
- Keep only principal components with enough explained variance

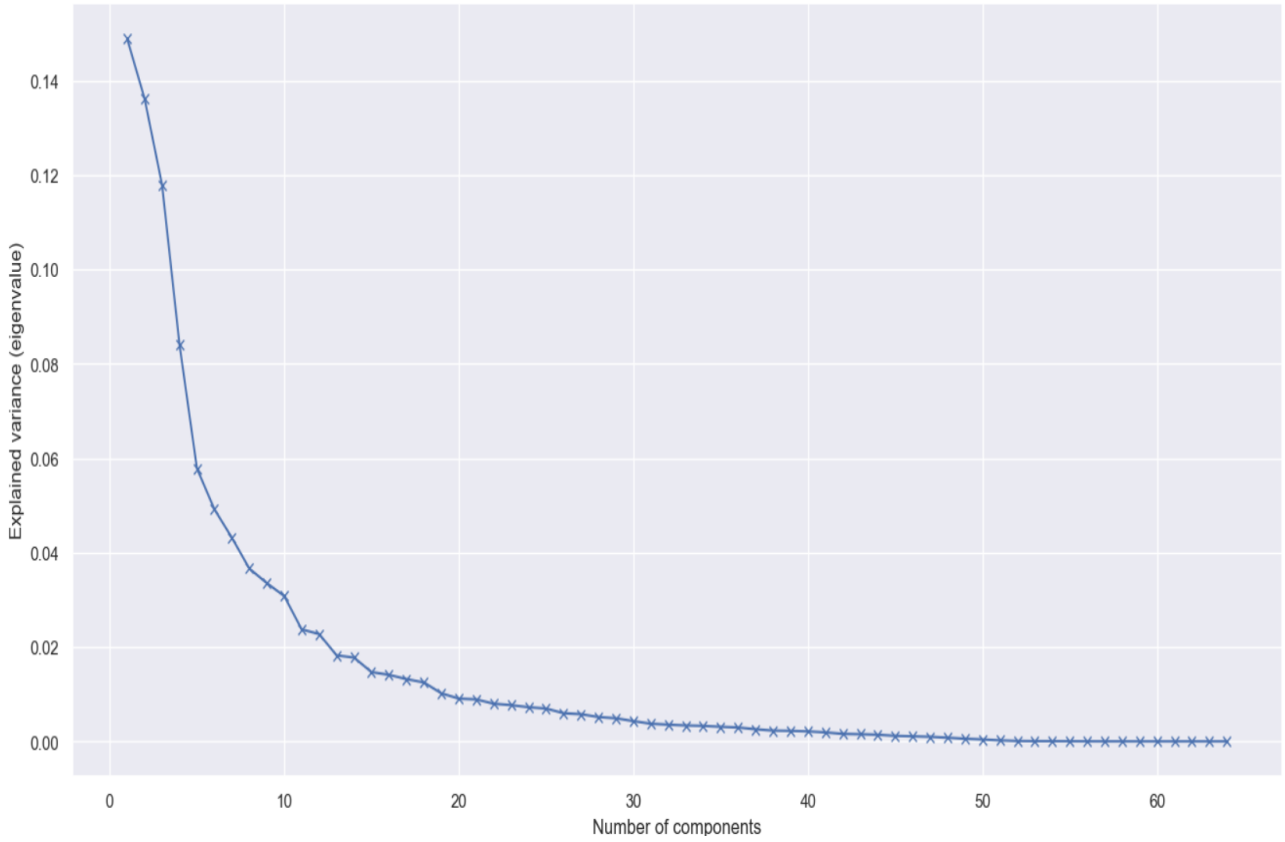


# Eigenvalues

- Eigenvalues tell us how much variance is explained by a principal component
- High eigenvalue = high variance
- High variance = lots of **useful** information
- Every principal component has its own eigenvalue
- Rank them from biggest to smallest → choose how many you want to keep

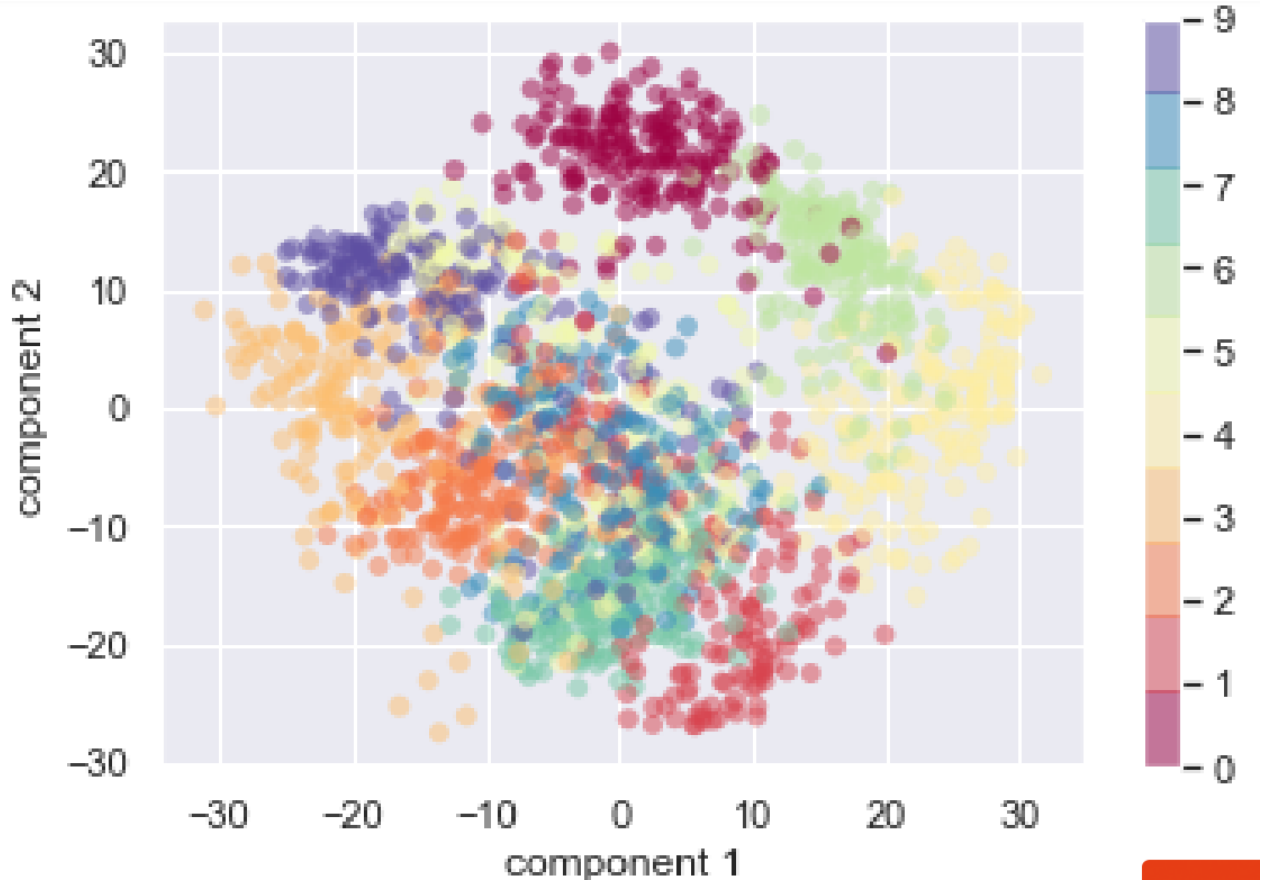
# Finding the ‘best’ amount of components

- Plot the eigenvalues in function of the number of components
- Does this graph look familiar?
- Elbow method!



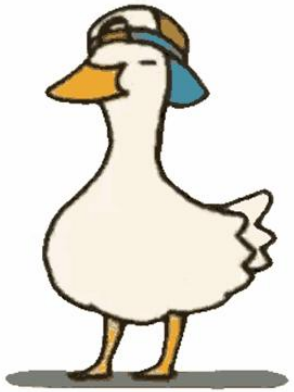
# Pusing PCA to the extreme

- What happens when we push PCA to the extreme?
- Take the digits dataset
- Reduce it to only 2 components
- Even now we can still make out the different numbers!
- Even with a reduction in variables of 97%!
- This show just how powerful PCA is



# Overfitting

- Since it is a dimensionality reduction technique, overfitting is not really a thing
- However, make sure you don’t reduce your data too much or you might lose valuable information!
- Same as with clustering, just use the elbow method and you’ll be fine :D



# Other uses of PCA

- Dimensionality reduction
  - Feature extraction
  - Data compression
- Data visualization
  - See digits dataset in two dimensions
- Noise reduction
  - By only keeping the principal components with high variance, we ‘filter’ out noise

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*Let's jump into action*

# Excercises

# First, let's explore this together

- Go to canvas and download the notebook "DSF 09 – Dimensionality Reduction Notebook"
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Follow along!



# Now it's up to you!

- Go to canvas and download the notebook "DSF 09 – Dimensionality Reduction Exercise"
- This time, no csv file is needed
- The data needed is retrieved in the notebook itself :)
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Go through the notebook and make the excercises




**I don't bite, ask me any questions you have!**